

Additive Manufacturing of Orthoses Can Improve Patient Outcomes

Adam Emile Aske, Viktoria Huszar, Yang Cui Solås

Dept. Biomedical Engineering, Oslo Metropolitan University, Oslo, Norway

adask2429@oslomet.no

s351963@oslomet.no

Abstract—Compliance monitoring holds significant importance in the evaluation of orthotic devices' efficacy, with wear-time information serving as a crucial parameter. Traditional orthoses are typically created from thermoplastic molds, a process that demands expertise and is time-consuming. However, additive manufacturing brings forth the potential for orthoses that are stronger, lighter, and easier to manufacture. This paper details the development and testing of a sensor-enabled static wrist orthosis designed for precise compliance monitoring, thus addressing the challenges of traditional orthosis creation and advancing the field towards improved patient outcomes.

Keywords—Orthosis; Compliance monitoring; Sensors; Additive manufacturing;

I. INTRODUCTION

Orthoses assists the rehabilitation of a variety of musculoskeletal impairments. From broken bones to post-stroke neurological damage, orthoses are a valuable and effective tool in many types of interventions. Objective measurements for how long an orthosis have been worn, compliance, is crucial to identify the efficacy of interventions. Correct compliance information may be unavailable in some patient populations. This article the development of a compliance monitoring orthosis. This article documents the development and testing of a sensor-enabled static wrist hand orthosis.

II. BACKGROUND

A. 3D-printed Orthoses

Traditionally, orthosis are made using thermoplastic molds. This is a time-consuming process and requires skilled personnel to manufacture. With the rise of additive manufacturing (3D-printing) technology, 3D-printed orthoses are rapidly being adopted as a viable alternative to conventional methods of construction[1].

Demeco et al. [2] identifies several problems with traditionally constructed orthoses: ventilation, hygiene, breathability, weight, moisture, lack of water proofing. All these factors may contribute to non-compliance in a patient, and thus worse patient outcomes. Several of the studies reviewed by Choo et al. [3] identifies 4 main attributes to explain why satisfaction with 3D-printed orthoses were higher than those made traditionally. Firstly, the contact surface is better aligned with the shape of the upper limb. Secondly, the strength of several

filament materials for 3D-printing are stronger and more versatile than traditional molds. Thirdly, a better distribution of pressure results in a more comfortable orthosis. Lastly, 3D-printed orthoses can be lighter, and thus more comfortable.

Oud et al. [1] notes that studies using 3D-printed orthoses are often conducted for a shorter amount of time than usual for long-term orthopedic interventions. The faster development and iteration speed 3D-printing may cause the results to favor 3D-printing because of the short time spans of studied interventions. In a long-term intervention these effects may be smaller or negligible. Additionally, rehabilitation using 3D printed orthoses was found to vary greatly in both frequency and volume, normally 3-6 times per week where patients wore an orthosis for 30 minutes to 6 hours, and the functional outcome was assessed using a variety scale such the Fugel-Meyer assessment, manual function tests, and block tests. Standardized testing paradigms to accurately determine efficacy of 3D printed orthoses.

B. Filament Material

The most popular materials used for 3D printed orthoses are thermoplastic polyurethane (TPU) and polylactic acid (PLA). PLA is a very widely used material due to features as non-toxic and biodegradable.[4] It is also known for its ease of use and being cheap, but they have a serious disadvantage as low temperature resistance. Temperatures around 50–60 °C may cause it to soften and thus deform [5]. Other characteristics of PLA material are rigidity and brittleness, whereas TPU is a semi-flexible material. TPU features excellent mechanical resistance to abrasion, impact, tearing, and chemical resistance to oils and greases. Furthermore, TPU is biocompatible and popular in the medical field. [6] Another biocompatible material is polyethylene terephthalate glycol (PETG) [7]. It is easy to print and possesses high transparency, impact resistance, chemical resistance, and weather resistance[8]. Worth to note that when they are reinforced with carbon fibers, they become stronger and more elastic [9].

Other available materials are acrylonitrile butadiene styrene (ABS) and nylon. Compared to PLA, ABS has

higher impact resistance. They have good thermal stability and can withstand any temperature encountered in daily use. However, ABS is not biodegradable which makes it less environmentally friendly. [5]. Nylon, also called polyamide, is stronger and more durable compared to ABS and PLA. Other advantages include being lightweight, highly flexible, wear-resistant, and not easily broken. [10]. Nylon is a relatively environmentally friendly and expensive material with a higher production difficulty. As an elastic material, it can resist high temperatures and chemical elements, and it also has high bending strength, impact, and scratch resistance. Its ability to bear dynamic loads makes it a recommended material for wrist-hand orthoses. [5]. Both TPU and nylon are good choices, but considering its good biocompatibility, flexibility and elasticity, wear resistance and comfort, TPU is ultimately chosen.

C. Compliance Monitoring

Devanad and Kedgley [12] investigated different methods of compliance monitoring in the literature. They identified 23 studies using sensors compliance monitoring: 6 used temperature sensors, 4 used step counters (pedometers), 6 used accelerometers, and 6 used pressure sensors. A few studies used a combination of sensors.

Temperature sensors achieve high accuracy, however climate and warm clothes can interfere with the results. Pressure sensors can achieve good results, but the calibration of thresholds is prone to errors. Step counters have the major drawback of not detecting time spent neither running nor walking. Accelerometers have high accuracy, but may give false positives during travel and other movements.

Thus, using a combination of temperature sensor and accelerometer a robust system for compliance monitoring may be possible. The sensors have complementary advantages and disadvantages. The accelerometer can give false positives during travel, but the temperature can invalidate those results. The temperature sensor can give false results when heavily clothed and climate changes, the accelerometer invalidate those results. It is important to note that the objective measurements of compliance may not indicate a patient's non-compliance; in many patient situations the patient themselves is not responsible for the compliance. Additionally, some patients may find the measurements intrusive, thus hindering compliance.

D. Lecture by Head of OT Department at Sunaas Hospital

The authors participated in a lecture about orthoses by the head of the Occupational Therapy department at Sunaas Rehabilitation Hospital. The lecture introduced the field of orthoses and gave insight into practicing clinician's experience working with them. The following statements are some of the authors' key takeaways from the lecture.

Hand function is an important part of quality of life and being able to participate in society.

A dynamic orthosis has some sort of active helping function that aids some sort of hand function.

A non-3D printed orthosis takes a lot of time and skill to get right and is difficult to alter once it's made. There can be no "pressure" inside the orthosis because a painful orthosis may disturb sleep and cause lesions.

An orthosis should preferably be visually appealing to increase patient compliance, especially for pediatric patients.

It would be beneficial if the orthosis assists the patient in using it correctly, a lot of people with this upper-limb impairment also have cognitive impairment. Improving upon an orthosis made at Sunaas can enable the occupational therapists to spend more time on patients rather than producing orthoses.

The clinicians need objective time tracking for the usage of orthoses. Assessing the efficacy of orthoses can be difficult without accurate information of how it has been used.

E. Previous Works

The previous students of the ACIT4730 course developed sensor-enabled orthoses. Three orthoses were developed, each incorporating a different type of sensor: temperature, accelerometer, and pressure. No testing procedures were documented.

III. METHODS

A. 3D Scanning

All authors' arm were scanned with an Einstar 3D Scanner (<https://www.einstar.com/products/einstar-3d-scanner?variant=43551575507118>). The scan with the least artifact was chosen for further development. The Einstar provided software, Exstar, was used to export the scanned arm as a 3D model (STL-file), depicted in Fig. 1.



Fig. 1. 3D-scanned arm.

B. 3D Design

MeshMixer (<https://meshmixer.com>) and Blender (<https://www.blender.org>) was used to process the 3D scan into an orthosis. Tools such as extrusion, Boolean, and subdivision were employed. The rendered orthosis is depicted in Fig. 2.

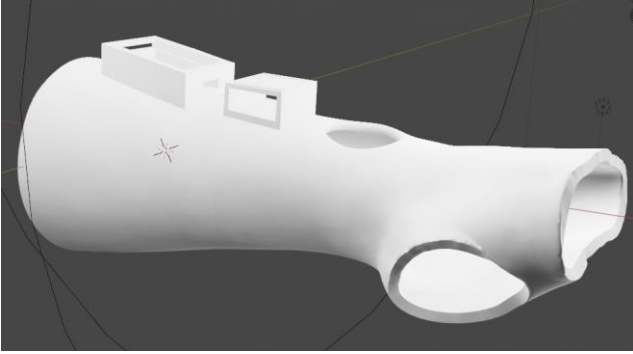


Fig. 2. Orthosis model.

C. 3D Printing

TPU is more complicated and time consuming to print compared to PLA, thus PLA was used for printing prototypes. The Bambu Lab X1-Carbon 3D printer (<https://eu.store.bambulab.com/en-no/products/x1-carbon-combo>) printed the PLA orthosis with parts. The orthosis is depicted in Fig. 3.



Fig. 3. Manufactured orthosis.

D. Sensors

- Seed Xiao ESP32s3 MCU w/ SD-card reader 32 GB on JAT connector
- Lithium-ion battery 600 Mah
- OLED screen 80x 160 (I2C)
- Infrared temperature sensor: MLX90614-BAA (I2C)
- IR receiver and IR transmitter (remote control) for

E. Monitoring

The microcontroller has two modes, sleep and awake, with different sampling rates. It enters awake mode if it records a measurement within the threshold values during sleep mode.

The infrared sensor uses emissivity and temperature, these are used to validate if the orthosis is worn.

Sleeping sample rate, awake sample rate, respectively, describe the sampling rate of the infrared sensor measurement.

F. Testing

The testing procedure aim is to find the optimal values for the parameters described in the previous subsection.

4 tests were conducted.

The first test was conducted in a lab environment with an idle temperature of 22 degrees. It was turned on 10 minutes before it was worn, then worn for 30 minutes, and 10 more idle minutes.

The second test was conducted in the streets of Oslo, Norway at an idle temperature of 10 degrees.

The third was started with the orthosis in

The fourth test started with the orthosis on the couch for 2 hours, then worn for 30 minutes, then placed back on the couch for 1 hour.

IV. RESULTS

The parameters used during the tests are described in table 1. The sleeping sample rate is described in sleep_fs. The awake sample rate is described in awake_fs. The emissivity values (calibration) and temperature thresholds are respectively described in emmisivity_t and temperature_t.

The result of each test is described in table 2.

TABLE I. TESTING REFERENCE

Test	Testing Parameters			
	Sleep_fs (Hz)	Awake_fs (Hz)	Emmisivity_t	Temperature_t (Celsius)
1	5	0.5	0.9-1	28
2	5	0.5	0.9-1	28
3	5	0.5	0.9-1	28
4	5	0.5	0.9-1	28

TABLE II.

Test	Testing Parameters			
	Total Test Time	Actual Wear Time	Mesaured Wear Time	Accu racy
1	50	30		
2	40			
3	260			
4				

How was it tested

-How did it perform in the tests

-How accurate the temperature sensor is in the setup: skin emission on the skin location (upper wrist), optimal distance between the skin and the sensor (~1-2 cm), sensor functioning in different temperature conditions (room , outdoor). Wetness of the skin. Comparing the sensor's accuracy with industrial IR temperature gun's accuracy.

-How reliable the software is to measure wearing time/ compliance (optimal threshold values, sampling based on RTC time).

- Battery consumption (testing the energy saving modes of the microcontroller e.g sensors, MCU operating in light/modem sleep mode).

- Testing network connectivity for getting precise time from NTP server

V. DISCUSSION

A. Orthosis

3D printing excels in the field of orthotics because it allows for personalized customization, thus enhancing comfort, fit, and effectiveness. Additionally, 3D printing supports rapid prototyping, enabling quick adjustments to orthotic parameters and designs based on testing feedback. 3D printing technology also facilitates advanced designs, producing orthoses that are both functional and aesthetically pleasing. Furthermore, a variety of available materials supports orthoses with different properties to meet various needs.

B. Compliance Monitoring

- How did it perform in the tests?
- What held it back ? What was good?
- How can be improved later?
- How the tests be improved?

C. Privacy & Security Assessment

As the device described in this article was only deployed in a lab environment and only testing data was recorded, security and privacy concerns are not considered. For real-life deployment of a device such as the one described in this article, it is important to consider several privacy and security concerns. The authors suggest an encryption scheme with a private key only the doctor has access to.

Additional risks include the lithium battery and exposed wires. The orthosis must be kept dry because the electronics and wires are exposed which brings the risk of electrocution. Waterproofing and impact protection must be considered. The electronics are powered by lithium-ion battery, a potential fire hazard. The battery management

system must be used to maximize battery performance and prevent potential risk of lithium-ion battery.

D. Bias

-How can it be improved ?

The device has 32 gb storage capacity, thus can record for a long time, the main problem is the battery life. Rechargeable and replaceable possibilities can be a good option to ensure proper functioning.

In addition, only four tests were conducted. More tests in various settings are needed for a more comprehensive understanding of the orthosis' performance.

VI. CONCLUSION

A. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE III. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^a. Sample of a Table footnote. (Table footnote)

Fig. 4. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

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